

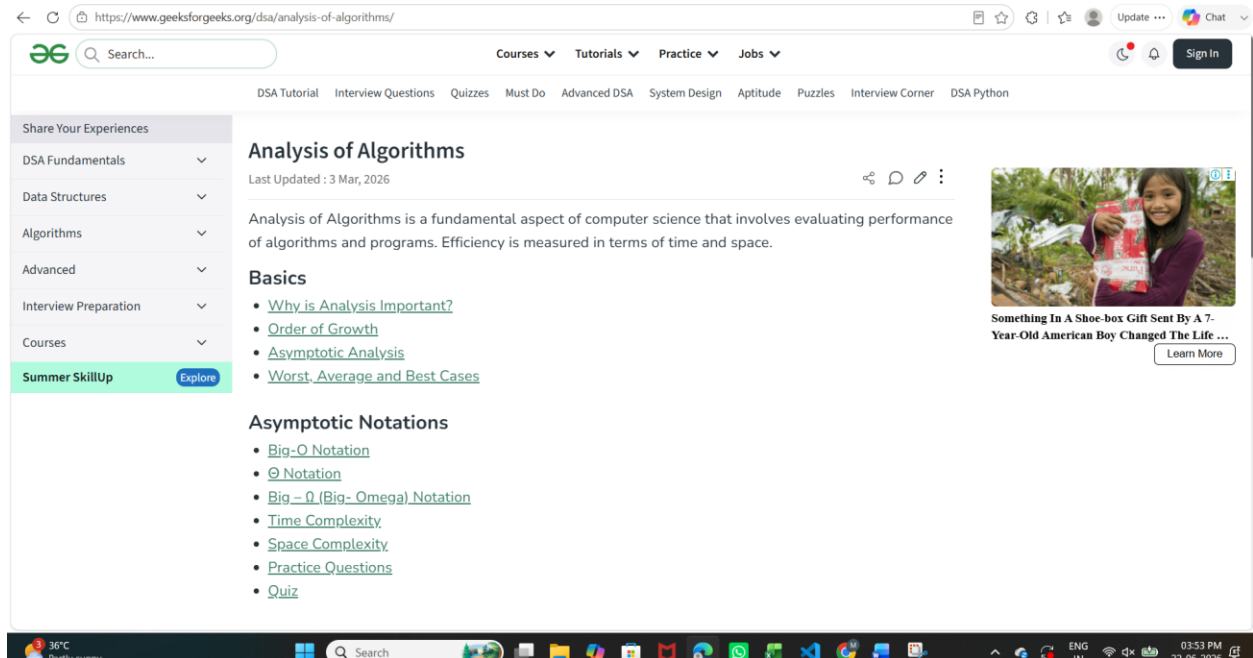
Module 2

Data Structures and Algorithms

Report on Self Learning:

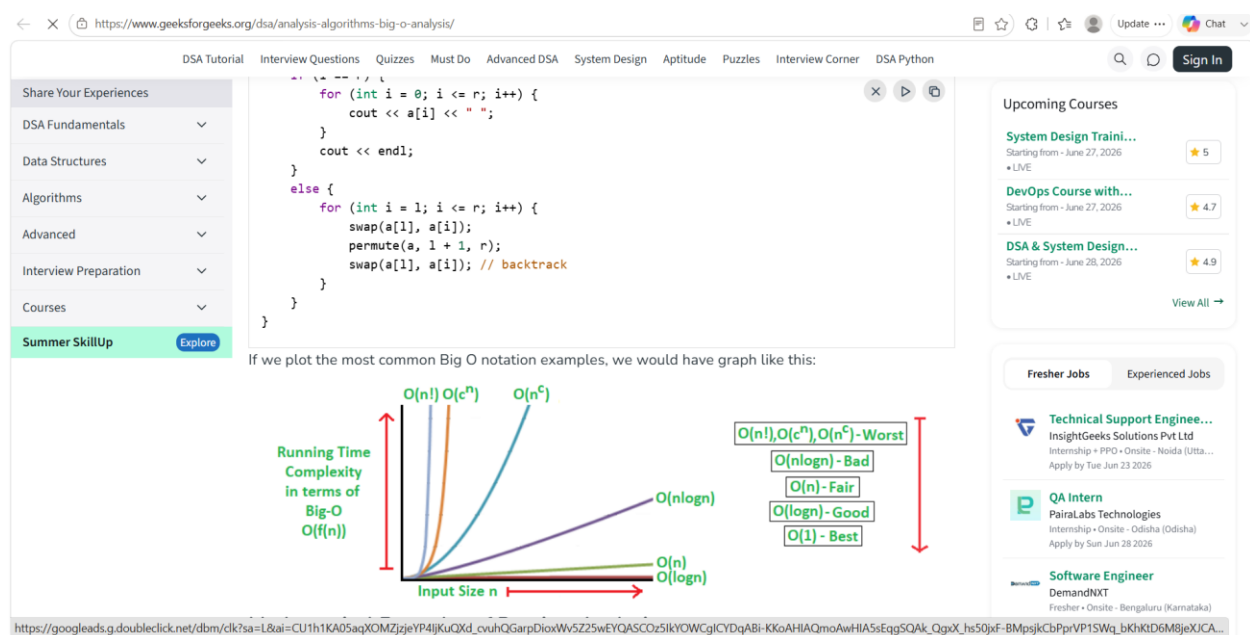
I Successfully Completed Module 2 – Data Structures and Algorithms and developed a thorough understanding of the basic concepts needed to design efficient and optimized software solutions. This module enabled me to appreciate the effective use of appropriate data structures and algorithms to organise, store, manage and process data. I discovered that choosing the correct data structures and algorithms is a crucial factor in optimizing software applications for performance, scalability, and efficiency.

This module covered the fundamental concepts of Data Structures and Algorithms, and their importance in designing efficient applications. I learned about various ways to organize and manipulate data, and how analysis of algorithms can be used to find the most efficient solution to a problem. I discussed the following subjects:



➤ Analysis of Algorithms – Understanding the introduction and importance of Data Structures and Algorithms, different types of data structures, and methods for analyzing algorithms. I became familiar with using frameworks to describe the behavior of algorithms and their efficiency.

➤ Time and Space Complexity – Understanding the performance of an algorithm in terms of time and space. I knew about asymptotic notations, including Big O notation, Big Omega notation, and Big Theta notation, to analyze the efficiency of algorithms. I also got to know the concepts of Best Case, Average Case and Worst Case analysis of algorithms and I have been able to analyse the time complexity of iterative and recursive methods.



➤ Arrays – Concept of arrays, how arrays are stored in memory and how to traverse, insert, delete and search an array. I learned what are arrays and why they are useful in storing multiple elements in contiguous memory locations.

← <https://www.geeksforgeeks.org/dsa/array-data-structure-guide/> Update ... Chat

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Array Data Structure

Last Updated : 3 Mar, 2026

An array is a fundamental and linear data structure that stores items at contiguous locations. Note that in case of C/C++ and Java-Primitive-Arrays, actual elements are stored at contiguous locations. And in case of Python, JS, Java-Non-Primitive, references are stored at contiguous locations. It offers mainly the following advantages over other data structures.

- **Random Access** : i-th item can be accessed in $O(1)$ Time as we have the base address and every item or reference is of same size.
- **Cache Friendliness** : Since items / references are stored at contiguous locations, we get the advantage of locality of reference.

Arrays are used to build other data structures like Stack Queue, Deque, Graph, Hash Table, etc. An array is not useful in places where we have operations like insert in the middle, delete from middle and search in a unsorted data.

Basics

- [Introduction](#)
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In Different Language

➤ **Linked Lists** – Learning about different types of linked lists such as Single Linked List, Circular Single Linked List, Doubly Linked List, and Circular Doubly Linked List. I learnt various operations like insertion, deletion, traversal and searching and their time complexity.

← <https://www.geeksforgeeks.org/java/linked-list-in-java/> Update ... Chat

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In order to create a LinkedList, we need to create an object of the LinkedList class. The LinkedList class consists of various constructors that allow the possible creation of the list. The following are the constructors available in this class:

- 1. LinkedList()**

This constructor is used to create an empty linked list. If we wish to create an empty LinkedList with the name list, then it can be created as:

```
LinkedList list = new LinkedList();
```

- 2. LinkedList(Collection C)**

This constructor is used to create an ordered list that contains all the elements of a specified collection, as returned by the collection's iterator. If we wish to create a LinkedList with the name list, then, it can be created as:

```
LinkedList list = new LinkedList(C);
```

Performing Different Operations on LinkedList

- 1. Adding Elements**

With the help of the add() method, we can add elements to a LinkedList This method can perform multiple operations based on different parameters. They are:

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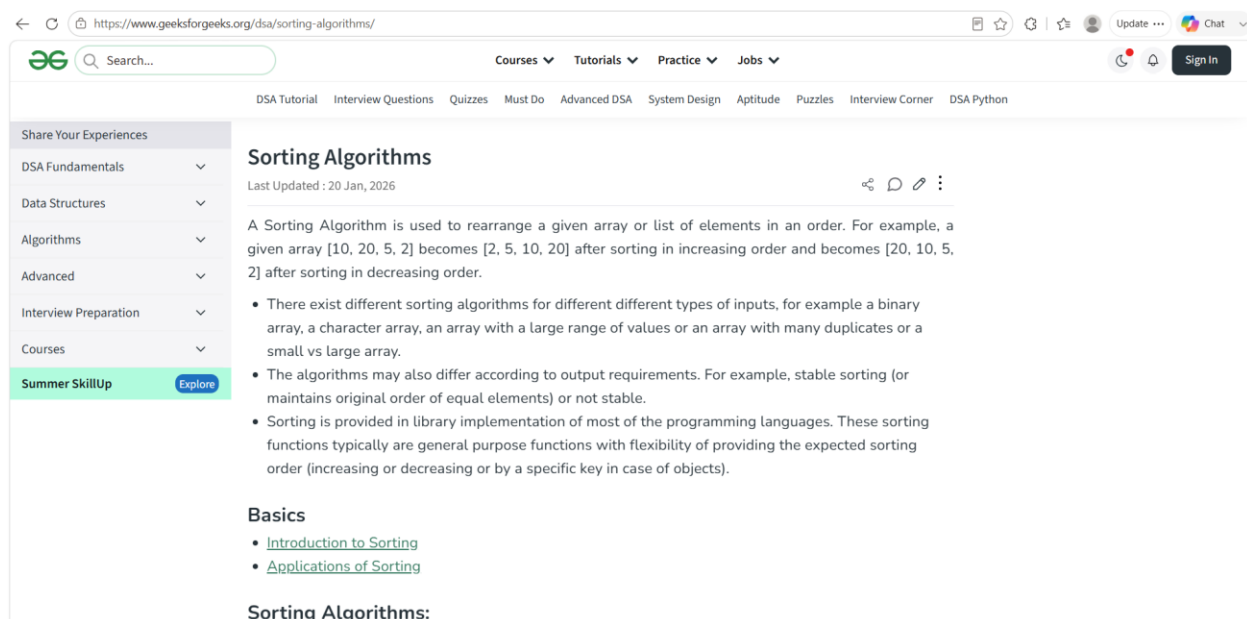
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➤ Searching Algorithms – Knowledge and Implementation of Searching Algorithms like Linear Search and Binary Search. I got to know about Linear Search and Binary Search, and how the latter is more efficient when the data is sorted.

➤ Sorting Algorithms – Learning and implementing various sorting methods like Bubble Sort, Insertion Sort, Heap Sort, Quick Sort and Merge Sort. I also studied their Best Case, Average Case and Worst Case complexities and compared their performance in various situations.



Completed practical coding exercises and implementation examples to gain insight into the use of these data structures and algorithms in real world scenarios. I was able to learn how to select appropriate solutions to problems by considering the needs and limitations of an application through practical activities. I gained experience in implementing different operations and analyzing the efficiency of the developed solutions.

I also learned problem solving skills and logical thinking in designing software applications. I have gained knowledge of the importance of using appropriate data structures and optimizing algorithms to enhance the efficiency of application execution, resource usage, and performance.

By engaging in coding practice and problem-solving exercises, I enhanced my analytical thinking skills and strengthened my ability to develop efficient and scalable solutions. This module has also given me a better insight into writing optimized code and finding alternative solutions to computational problems.

To conclude, this module has helped me immensely in my knowledge of Data Structures and Algorithms and also enhanced my problem-solving skills. It gave me the ability to write efficient, maintainable and scalable software applications and enhanced coding efficiency and performance.